RADIANDA SETIAWAN

AI Engineer | Full Stack Software Engineer

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PROFILE SUMMARY

Second-year Computer Science student (5th semester) at Bina Nusantara University with hands-on experience building full-stack systems, ERP/Supply Chain integrations, and applied machine learning research. Proficient in C#/ASP.NET, React, and Python, with experience in scalable backend architecture (REST APIs, cloud deployment on Microsoft Azure) in legacy-system migration and RAG-based WhatsApp chatbots. Strong interest in AI/Intelligent Systems, demonstrated through a research paper accepted at an international conference (COMPGINEER 2026) and hands-on projects spanning NLP, computer vision, and predictive modeling. Seeking an AI Engineer / Software Engineer role to contribute to data-driven and AI-powered products.

WORK EXPERIENCE

Software Engineer — PT Batavia Satu (Part-time) *Jun 2026 – Present | West Jakarta, Hybrid*

- Built a WhatsApp chatbot for a skin and laser clinic, integrated with the company SQL Server database per proper SOP, using RAG with vector embeddings; customers and doctors can schedule or free up appointments through it.
- Administered company servers: set up local MCP (Model Context Protocol) connectivity on the company server.
- Contributed to the completed migration of a clinic network's internal system (Batavia Central) from VB.NET WebForms/DevExpress to ASP.NET Core MVC with Dapper and SQL Server (210 pages, 17 modules, excl. BI);delivering CRM modules: Loyalty, Redemption, E-Voucher, Service Packages, SMS, Photo Gallery, Purchase History.
- Implemented write actions with transactions, audit trail, and soft delete; matched PDF/Excel/CSV exports to legacy; applied anti-forgery tokens, upload validation, and parameterized queries.
- Analyzed legacy VB.NET code to decide whether to migrate, fix, or disable each behavior, fixing legacy bugs (e.g., duplicated refund log rows, invalid SAP posting SQL); also migrated Finance pages (FIFO, SAP Extract/Export) and published code from DevOps to the servers.
- Built a schema-resilient data layer for missing tables/columns across environments, and UI- and server-side safety switches that disable high-risk stock actions (Get COGS, Post to SAP) pending Finance confirmation.
- Architected and developed end-to-end B2B order and fulfillment workflows for a Supply Chain Management & ERP system, building modules for Purchase Orders, Sales Orders, Delivery Orders, Invoicing, and real-time inventory tracking.
- Engineered centralized messaging integrations with WeChat and WhatsApp using C#, ASP.NET Core, and LINQ; rapidly built and validated full-stack prototypes using React, ASP.NET Core APIs, SQL Server (SSMS), and Microsoft Azure.
- Built an end-to-end automated web scraping system using Python, delivering extracted data via high-performance FastAPI backends and interactive React frontends, with persistence in SQL Server and deployment on Microsoft Azure.
- Designed robust RESTful APIs with Data Transfer Object (DTO) mappings, implementing efficient relational database schemas and optimized queries in SQL Server.
- Configured, deployed, and managed full-stack backend services, background workers, and API endpoints on Microsoft Azure App Services.

Software Engineer (Internship) — PT Cellbox *Jul 2025 – Aug 2025 | West Jakarta, On-site*

- Built and maintained end-to-end web applications implementing the Model-View-Controller (MVC) architecture across frontend, backend, and relational database layers.
- Leveraged Dapper micro-ORM to execute lightweight, optimized object mapping and high-throughput SQL queries against the database.
- Designed, implemented, and documented robust REST APIs supporting complete CRUD lifecycles for business-critical entity management.
- Structured normalized relational database schemas, authored complex queries, and ensured data integrity across interconnected system modules.

Administrator (Part-time) — PT Cipta Duta Mahakarya *May 2024 – Aug 2024 | Batam, On-site*

- Managed end-to-end billing workflows, including drafting, verifying, and reconciling invoices with delivery orders to ensure accuracy across all logistics transactions.
- Served as the primary operational liaison, actively communicating with clients, distributors, and delivery couriers to provide shipment updates and resolve fulfillment inquiries.
- Conducted on-site visits to directly engage with clients and inspect cargo handling; monitored warehouse stock levels and performed routine physical inventory audits.
- Coordinated multi-channel dispatch schedules between warehouses and third-party couriers to streamline route efficiency and minimize delivery lead times.

RESEARCH & PUBLICATIONS

Conference Paper — Accepted at COMPGINEER 2026 — "Evaluating the Financial Impact of Cost-Sensitive Random Forest and SMOTE-Tomek" *Apr 2026 – Aug 2026*

- Authored and secured acceptance for a machine learning research paper to be presented at COMPGINEER 2026, focusing on mitigating extreme class imbalance and minimizing real-world financial losses in credit card fraud detection.
- Implemented hybrid data resampling (SMOTE-Tomek): applied SMOTE to synthesize minority fraud cases and Tomek Links to clean overlapping decision boundary noise in highly imbalanced transactional datasets.
- Engineered a Cost-Sensitive Random Forest classifier by applying custom penalty matrices, heavily penalizing false negatives to prioritize detection of actual financial theft over standard global accuracy.
- Fine-tuned penalty weights to achieve an optimal balance, significantly reducing undetected fraud cases and lowering overall estimated financial loss for simulated banking operations.
- Benchmarked the integrated architecture against standard baselines using PR-AUC, minority recall, false negative rate, and projected monetary impact.

SELECTED PROJECTS

Smart Hoax Detector — AI-Powered Political Misinformation Classification — *NLP · Group project, BINUS University*

- Architected a comparative ablation study benchmarking 5 models (SVM, 1D-CNN, BiLSTM+Attention, XLM-RoBERTa, IndoBERT V2); the fine-tuned IndoBERT model achieved a peak Macro F1-score of 99.39% decoding Indonesian stylometric variations.
- Processed a dataset of 30,000+ articles from 4 sources using strict downsampling and regex-based structural cleansing to remove media watermarks and journalistic footprints, preventing algorithmic bias.
- Integrated a Hybrid RAG (Retrieval-Augmented Generation) pipeline with DuckDuckGo web scraping for real-time fact-checking, with an automated fallback to the Wikipedia API during rate-limiting events.
- Built a decoupled cloud architecture with a React.js frontend (Vercel) and an asynchronous FastAPI inference backend (Hugging Face Spaces), automated via GitHub Actions CI/CD.

Tech: *Python, IndoBERT, XLM-RoBERTa, RAG, FastAPI, React.js, Hugging Face*

Driver Drowsiness Detection System (ADAS) — Real-Time Computer Vision — *Computer Vision · Group project, BINUS University*

- Implemented Dlib's 68-point facial landmark predictor (HOG + Linear SVM) to compute real-time Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) for detecting prolonged eye closure and yawning.
- Ran a multi-dimensional Grid Search over EAR, MAR, and temporal-frame thresholds, evaluated on the UTA Real-Life Drowsiness Dataset (141 videos) under a strict "Maximum Recall" paradigm, achieving 83% recall on critical drowsiness states.
- Achieved real-time inference at 31.11 FPS on a local x86 machine without a GPU by using deterministic classical computer vision instead of deep neural networks, for full algorithmic interpretability.
- Built a full-stack app with a React.js frontend and Flask backend using Base64-encoded frame streaming and multi-state visual/audio alerts, deployed via GitHub Actions to Vercel and Hugging Face Spaces.

Tech: *Python, Dlib, OpenCV, Flask, React.js, GitHub Actions*

LQ45 Market Direction Predictor — *Machine Learning · Group project, BINUS University*

- Engineered a Support Vector Classifier (SVC) with an RBF kernel, tuned via GridSearchCV and 5-fold cross-validation to optimize the C and gamma parameters for non-linear market pattern recognition.
- Processed historical market data into 14 technical indicators (EMA, RSI, MACD, Bollinger Bands) with strict sequential data splitting and isolated feature scaling to prevent data leakage and preserve time-series integrity.
- Built a decoupled architecture with a React.js frontend (Vercel) and a containerized Flask backend (Docker) on Hugging Face Spaces, integrated with the Yahoo Finance API for live market data.
- Established CI/CD and MLOps automation using GitHub Actions with DVC and MLflow concepts, and validated the system through usability testing with 5 student traders.

Tech: *Python, Scikit-Learn, SVC, Flask, Docker, React.js, yfinance*

Bookuger — Full-Stack Library Management System & O2O Architecture — *Software Engineering · Group project, BINUS University*

- Built a mobile-first, responsive React.js (Vite) interface with real-time catalog filtering and dynamically generated Digital Member QR Codes for frictionless, contactless checkout.
- Engineered a secure RESTful API in C#/.NET Core with stateless JWT-based Role-Based Access Control (RBAC), and automated late-fine calculations with a member blacklisting system for disciplinary enforcement.
- Interfaced the admin dashboard with external HID optical/barcode scanners for high-speed checkout verification, and enforced pessimistic concurrency control on an Azure SQL Database (Entity Framework Core) to prevent race conditions.

- Documented the system with a 98-page Software Requirements Specification (SRS), including UML diagrams and an ERD, and deployed via automated CI/CD: frontend on Vercel, backend API on Azure App Service.
  Tech: *C#, ASP.NET Core, Entity Framework Core, Azure SQL, React.js (Vite), JWT*

Omnichannel Messaging & Webhook Integration Service — *Web Development · PT Cellbox Internship*

- Built robust RESTful API endpoints in ASP.NET Core (C#) to securely receive, process, and route incoming webhook events from the WeChat Sandbox and WhatsApp platforms.
- Architected the database schema using Microsoft SQL Server and Entity Framework Core, with efficient LINQ queries and Data Transfer Object (DTO) mappings for clean data transitions between external APIs and internal storage.
- Engineered backend logic for message parsing, contact resolution, and status tracking (read receipts, unread message counters), laying the groundwork for real-time bi-directional chat interfaces.
- Configured and deployed the backend services to Microsoft Azure App Services, ensuring high availability, secure endpoints, and continuous integration; later merged into a larger enterprise management system.
  Tech: *C#, ASP.NET Core, Entity Framework Core, SQL Server, Azure App Services*

Autonomous AI-Powered E-Commerce Scraping Engine — *Artificial Intelligence · Personal / PT Batavia Satu*

- Integrated the Google Gemini API to parse natural-language scraping intents and autonomously filter raw scraped data, ensuring the SQL database is populated only with highly relevant product variations.
- Bypassed Datadome anti-bot protection using Playwright with persistent contexts, human-like scrolling intervals, and randomized mouse movements along cubic Bezier curves to maintain natural session fingerprints.
- Developed an autonomous CAPTCHA solver for soft-block slider puzzles using OpenCV, applying Canny edge detection and template matching to calculate precise X-axis distances for automated drag-and-drop.
- Built a fault-tolerant "Emergency Stop" system that gracefully terminates the script upon detecting unresolvable hard blocks, with automated SMTP email alerts for team intervention.
- Wrapped the engine in a FastAPI RESTful backend with upsert logic via pyodbc to store dynamic product data (variants, prices, ratings, stock) in a Microsoft SQL Server database.
  Tech: *Python, Google Gemini API, Playwright, OpenCV, FastAPI, SQL Server, pyodbc*

EDUCATION

Bachelor of Computer Science — Bina Nusantara University (BINUS), Alam Sutera *2024 – Present (5th Semester)*

- Specialization: Intelligent System Streaming.
- Active in student organizations and national-level technology competitions.

Science Graduate — Sekolah Mondial, Batam, Kepulauan Riau *2021 – 2024*

LICENSES & CERTIFICATIONS

- Microsoft AI-900T00-A — Learning AI from the Basics, Microsoft (Mar 2026).
- Internship Certificate, PT Cellbox (Jul 2025).

SKILLS

Programming Languages: Python, C#, VB.NET (legacy code analysis), JavaScript, TypeScript, Java, C, SQL (T-SQL)

AI & Machine Learning: Scikit-Learn, Pandas, NumPy, OpenCV, Dlib, PyTorch, TensorFlow, Hugging Face Products, Transformer Architectures, IndoBERT, XLM-RoBERTa, BERT (Language Model), Random Forest, SMOTE / SMOTE-Tomek, Cost-Sensitive Learning, Principal Component Analysis, Fraud Detection, Retrieval-Augmented Generation (RAG), Google Gemini API (LLM), Vector Embeddings, Model Context Protocol (MCP), Computer Vision, NLP, Data Analysis, Data Engineering, Feature Engineering, Academic Research, Academic Writing

Web & Backend: React.js, Next.js, Node.js, Tailwind CSS, Vite, ASP.NET Core (MVC & Web API), Razor, AngularJS, Entity Framework (EF) Core, Object-Relational Mapping (ORM), Dapper ORM, LINQ, FastAPI, Flask, REST API Design, DTO Mapping, Webhook Integration, WeChat API, WhatsApp API, Chatbot Development, E-Commerce

Database & Cloud: SQL Server (SSMS), Database Integration, MySQL, PostgreSQL, Supabase, Neon (Serverless Postgres), Microsoft Azure (App Services, DevOps), Cloud Deployment, Server Administration, Vercel, Hugging Face Spaces, Docker, GitHub Actions, CI/CD, Git, GitHub, GitLab

Automation & Tools: Playwright, Visual Studio, TFVC, Claude Code, System Resilience & Fault Tolerance, Web Mapping, Computer Hardware

Business & Domain: ERP Systems, Supply Chain Management, Logistics Management, Inventory Management, Invoicing, Client Relations, Legacy System Migration